# MCQA

## Minecraft answers you can check

Cited Wiki evidence and official recipe calculations

Fully local · Minecraft Java 26.3 target

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# Players need facts and exact amounts

## Gameplay questions

Wiki pages mix editions and historical rules.

Retrieve passages and show revision links.

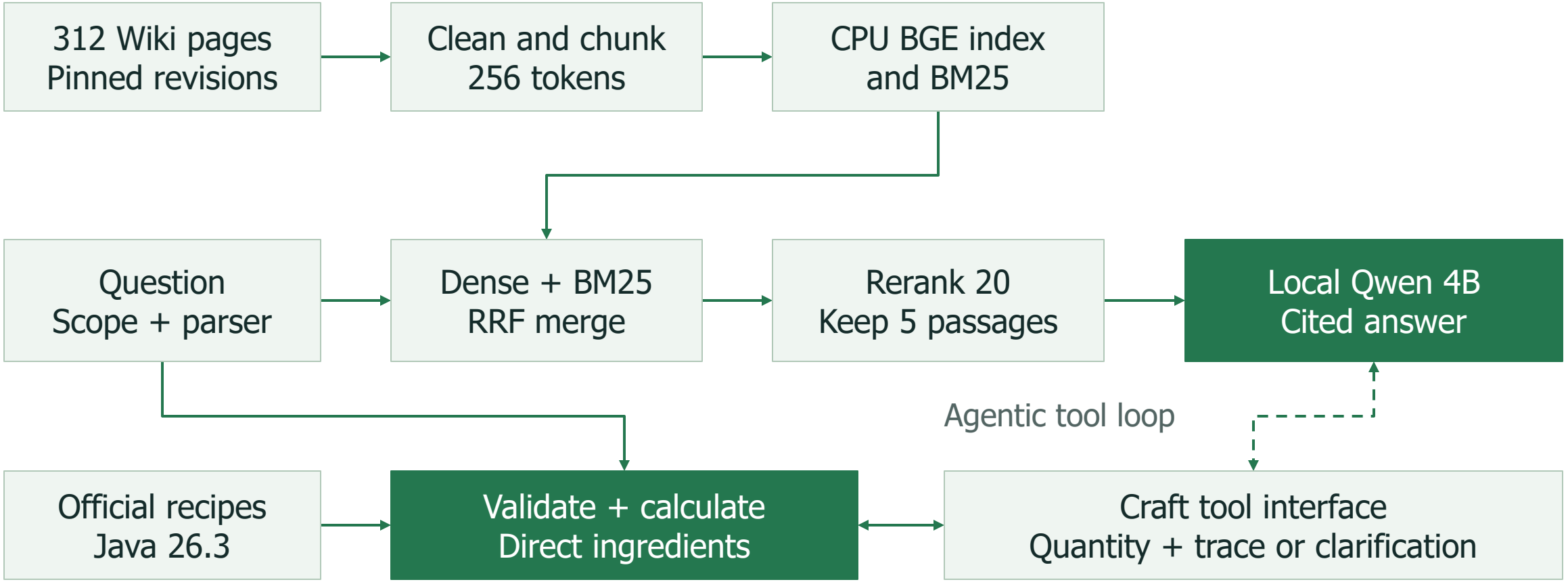
## Crafting questions

"How many iron ingots for 3 hoppers?"

Read an official recipe and calculate the batches.

Scope: direct crafting and supported Java evidence. Other editions need a clear boundary.

# Two answer paths, one local system




Clear recipes bypass generation

Embedding and reranking run on CPU. Generation uses local Ollama.

# Corpus and chunking

**312**

distinct real Wiki pages

About 817,000 words
Exact page revisions retained

## 256-token chunks

Keep section and paragraph boundaries.
Split oversized paragraphs.
No overlap.

Compared 128, 256 and 512 tokens.
Historical reranker page hits
were similar: 74.1% to 74.5%.

The synthetic Q&A pool chooses page titles and supports older diagnostics. It is not the retrieval corpus.

# Small models keep the system local

| Role | Choice | Purpose |
| --- | --- | --- |
| CPU embeddings | BGE-small-en-v1.5 | 384-dim semantic vectors |
| Hybrid search | BM25 + dense, RRF | Match names and meaning |
| CPU reranking | MS MARCO MiniLM-L6-v2 | Joint passage scoring |
| Local generation | Qwen3 4B Instruct | Citations and tool calls |

Retrieve 50 per branch. Fuse ranks. Rerank 20. Answer from the top 5.

Live demo next: recipe quantity, cited gameplay answer, edition boundary.

# Retrieved evidence improves factual answers

| Variant | Judged correct | Quote in top 5 |
|---|---|---|
| No retrieval | 9/34 | Not applicable |
| Dense | 27/34 | 31/34 |
| Hybrid + reranker | 26/34 | 30/34 |

**Evidence gives the model facts it could otherwise guess.**

Reranking did not establish a gain over dense retrieval.

Frozen v1 test: 34 facts + 6 negatives. Automatic same-model judge. Historical implementation.

# The recipe tool improves quantity answers

| Same 40 questions | Exact quantity | Correct (%) |
|---|---|---|
| Without recipe tool | 1/40 | 2.5% |
| With recipe tool | 37/40 | 92.5% |

**Official recipe slots determine ingredient consumption.**

All 3 misses made no tool call, so routing is the weak point.

Frozen v2 test. Direct ingredients only, without inventory reuse or recursive material planning.

# Current quality needs two kinds of checks

| Current variant | LLM judge correct | BERTScore F1 |
|---|---|---|
| Dense | 6/8 | 0.654 |
| Hybrid + reranker | 7/8 | 0.600 |

**Correctness checks facts. BERTScore checks similarity.**

Both compare with pinned references. Neither is human validation.

Latest separate study: 8 test facts. Small sample, same-model judge. No general accuracy gain established.

# Correct numbers can still hide wrong claims

| Observed failure | Why it happens |
| --- | --- |
| Useful Java evidence removed | Historical and edition rules share paragraphs |
| Right ingredient, wrong explanation | The model adds claims beyond the tool result |
| Judge verdict can be wrong | Generator and judge use the same local model |

**Keep raw answers, citations and tool traces for review.**

Next: improve mixed-section handling and independently review labels on a fresh test set.

# AI audit: three decisions and one caught error

## Three AI-owned decisions

256-token section chunks: balance context and cost.

CPU BGE embeddings: compact English retrieval.

Official recipe calculation: exact quantities, with a tool loop.

## One actual error catch

The old scorer credited a refusal that repeated a number.
Fix: check the requested ingredient and reject refusals.

Claude Code and Codex built and checked the system.